

Discrete and Continuous Methods for Packing Problems

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Preface

This book develops a unified framework for packing problems, combining discrete optimization, graph theory, and continuous methods from applied mathematics.

Chapter 1

Optimal Packing of Non-Convex Polygons

This project is an exercise in optimization and High performance computing. It is inspired in the classical problem of sphere packing with a twist: We seek to find the smallest square container that can fit N identical non-convex polygons without overlap.

1.1 Mathematical Formulation

We consider a set of N identical non-convex polygons

$$\mathcal{P} = \{P_1, \dots, P_N\},$$

Each polygon can be described with the state variables $(x_i, y_i, \theta_i) \in \mathbb{R}^2 \times [0, 2\pi)$.

Definition 1.1 (Feasible Configuration). A configuration $\mathbf{c}$ is feasible if:

1. All vertices lie within the square domain $[0, L]^2$.
2. No two polygons overlap.

The objective is to find the minimal L such that all polygons are placed inside the square without overlap. Formally, we seek to solve:

$$\begin{aligned} \min_{L, \mathbf{c}} \quad & L \\ \text{s.t.} \quad & \mathbf{c} \text{ is a feasible configuration of } \mathcal{P} \text{ in } [0, L]^2. \end{aligned}$$

1.1.1 Feasibility-Based Search

A common strategy is to reduce the optimization problem to a sequence of feasibility tests. For a fixed container size L , one asks whether there exists a configuration $\mathbf{c}$ such that all polygons fit

inside $[0, L]^2$ without overlap. This leads to the decision problem

Find $\mathbf{c}$ such that $\mathbf{c}$ is feasible in $[0, L]^2$.

By performing a search over values of L . Using bisection, one can approximate the minimal feasible container size. This approach shifts the difficulty from direct optimization to repeated feasibility detection in a highly constrained configuration space.

1.1.2 Energy-Based Formulations

An alternative and widely used approach replaces hard geometric constraints with soft penalties by introducing an energy function of the form

$$E(\mathbf{c}; L) = E_{\text{overlap}}(\mathbf{c}) + E_{\text{boundary}}(\mathbf{c}; L),$$

where overlaps and boundary violations contribute positive energy. In this formulation, feasible configurations correspond to zero-energy states.

This transforms the constrained optimization problem into an unconstrained (but non-convex) minimization problem over $\mathbb{R}^{2N} \times S^1$. The resulting energy landscape is typically rugged, containing many local minima corresponding to jammed or partially overlapping configurations.

Energy-based formulations naturally motivate the use of stochastic optimization techniques, such as simulated annealing and Monte Carlo methods. We use them to explore the configuration space and find feasible or near-feasible configurations, which are designed to explore complex non-convex landscapes.

Computational and algorithmic challenges. The packing problem studied in this work presents several fundamental difficulties, both geometric and algorithmic.

1. **Highly non-convex energy landscape.** The objective function associated with packing design by associating a penalties to overlaps and boundary violations is highly non-convex. Additionally, it exhibits a large number of local minima. As will be shown in the next chapter, even simplified model like the Ising model possesses a non-convex energy landscape. As a consequence, purely local optimization methods get trapped in suboptimal configurations, requiring use of global or stochastic optimization techniques.
2. **Geometric complexity due to non-convex shapes.** By the nature of non-convex polygons, the overlap checks require careful geometric computation, often involving decomposition into convex components or the evaluation of separating axes. This geometric complexity significantly increases the computational cost of evaluating candidate configurations.

3. **Exponential growth of the search space.** The configuration space grows exponentially with the number of objects N , as each polygon contributes translational and rotational degrees of freedom. Consequently, exhaustive search methods become infeasible even for moderately large systems.
4. **Computational intractability (NP-hardness).** The problem is computationally intractable in general. This follows from the simplified version being hard: if all polygons are restricted to be circles. This can be seen by considering the associated *decision problem*: given a set of circles with prescribed radii and a square container of fixed side length, determine whether all circles can be placed inside the square without overlap.

This decision problem is known as the *circle packing problem* and has been shown to be NP-complete. Since circle packing is a special case of the more general polygon packing problem, it follows that the general problem is NP-hard.

Chapter 2

Simulated Annealing and Its Parallel Variants

The energy-based formulation of Chapter 1 reduces polygon packing to minimizing a highly non-convex function $E(\mathbf{c}; L)$ over $\mathcal{C} = \mathbb{R}^{2N} \times [0, 2\pi)^N$. This chapter develops the stochastic optimization machinery used to explore that landscape. We begin by establishing the common probabilistic target, then derive the base Markov chain dynamics and the three parallel coordination strategies—MS, ER-MS, and PT—as distinct approximations to sampling from that target.

2.1 The Common Probabilistic Target

All three methods studied in this work can be understood as strategies for approximating samples from the *Gibbs measure* at temperature $T > 0$:

$$\pi_T(\mathbf{c}) = \frac{1}{Z(T)} \exp\left(-\frac{E(\mathbf{c}; L)}{T}\right), \quad Z(T) = \int_{\mathcal{C}} e^{-E(\mathbf{c}; L)/T} d\mathbf{c}.$$

As $T \rightarrow 0$, π_T concentrates on the global minimizers of E . Feasible configurations correspond to zero-energy states, so finding a feasible packing is equivalent to sampling from π_T in the low-temperature limit.

The methods differ in how they allocate R parallel workers to approximate this target:

- **MS** draws R independent samples from separate realizations of the annealing process.
- **ER-MS** couples workers via a periodic selection–mutation operator, forming an interacting particle system that biases mass toward low-energy regions.
- **PT** couples R chains thermodynamically via swap moves, targeting the *joint* Gibbs distribution over the replica ensemble.

2.2 Base Markov Chain Dynamics

2.2.1 Metropolis–Hastings Kernel

Given a configuration $\mathbf{c}_k$ at step k , a candidate $\mathbf{c}'$ is drawn from a symmetric Gaussian proposal kernel $q(\cdot \mid \mathbf{c}_k)$ and accepted with probability

$$\alpha(\mathbf{c}_k, \mathbf{c}') = \min\left(1, e^{-\Delta E/T_k}\right), \quad \Delta E = E(\mathbf{c}'; L) - E(\mathbf{c}_k; L).$$

Symmetry of q ensures the satisfaction of detailed balance with respect to π_{T_k} . We need detailed balance to guarantee convergence of our Markov chain to the target distribution. The resulting transition kernel is

$$P_k(\mathbf{c} \rightarrow d\mathbf{c}') = q(\mathbf{c}' \mid \mathbf{c}) \alpha(\mathbf{c}, \mathbf{c}') d\mathbf{c}' + r(\mathbf{c}) \mathbf{1}_{\mathbf{c}}(d\mathbf{c}'),$$

where $r(\mathbf{c}) = 1 - \int q(\mathbf{c}' \mid \mathbf{c}) \alpha(\mathbf{c}, \mathbf{c}') d\mathbf{c}'$ is the rejection probability.

Proposal structure. At each step, a polygon index $i \sim \text{Uniform}\{1, \dots, N\}$ is selected and a joint positional and rotational perturbation is applied:

$$x'_i = x_i + \delta_x, \quad y'_i = y_i + \delta_y, \quad \theta'_i = \theta_i + \delta_\theta,$$

with $(\delta_x, \delta_y)^\top \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 I_2)$ and $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, drawn independently.

2.2.2 Cooling Schedule and Mixing

We use a *geometric* cooling schedule:

$$T_k = T_{\max} \rho^k, \quad \rho = \left(\frac{T_{\min}}{T_{\max}}\right)^{1/K},$$

where K is the total number of steps in a bisection probe. The geometric schedule is characterized by a constant multiplicative temperature reduction ρ per step, which gives equal thermal exposure at each decade of energy: the number of steps spent in any interval $[cT, T]$ is $-\log c / \log(1/\rho)$, independent of the absolute temperature level.

As stated in the notes from Sokal, "Monte Carlo is an extremely bad method; it should be used only when all alternative methods are worse. " Convergence to the global minimum under the Metropolis–Hastings dynamics requires logarithmic cooling $T_k = c / \log(1 + k)$ for a constant c exceeding the depth of the deepest local energy barrier [1]. Such schedules are computationally infeasible for the problem sizes considered here; the geometric schedule instead prioritizes practical convergence within a fixed computational budget, with $T_{\min}$ and $T_{\max}$ fixed by pilot calibration (Section 2.5.6).

2.2.3 Integration with Bisection

Each method operates within an outer bisection loop over the container side length L . For a fixed bracket $[L_{\text{lo}}, L_{\text{hi}}]$, a probe evaluates the midpoint $L = (L_{\text{lo}} + L_{\text{hi}})/2$ by running K steps of the SA-based search. If a feasible configuration is found, the bracket contracts: $L_{\text{hi}} \leftarrow L$; otherwise $L_{\text{lo}} \leftarrow L$. The SA energy function $E(\mathbf{c}; L)$ is therefore re-evaluated at a different L on each bisection step, and the Gibbs target π_T changes accordingly.

2.3 Method A: Multi-Start Simulated Annealing (MS)

2.3.1 Definition

MS instantiates R independent Markov chains, each evolving under its own copy of the base SA kernel:

$$\mathbf{c}_k^{(r)} \xrightarrow{P_k, T_k} \mathbf{c}_{k+1}^{(r)}, \quad r = 1, \dots, R.$$

Workers share no state during a probe. Seeds are generated deterministically as

$$s_r = \text{splitmix64}(g \oplus \text{hash}(N, \text{run_id}, \text{probe_idx}, r)),$$

where g is the global seed, ensuring exact reproducibility.

2.3.2 Probabilistic Analysis

Let p denote the probability that a single SA chain finds a feasible configuration within the allotted probe time τ . The probability that at least one of R independent workers succeeds is

$$P_{\text{success}} = 1 - (1 - p)^R.$$

For small p , this grows as Rp to first order, so R workers provide a linear improvement in success probability at linear cost. More precisely, if the first-success time for a single chain is Geometric(p)-distributed with mean $1/p$, then the minimum across R independent chains has mean $1/(Rp)$: the expected time to first feasibility scales as $1/R$.

2.3.3 Limitations

The independence of MS chains is both its computational strength (zero communication overhead) and its fundamental weakness. Workers cannot exploit configurations discovered by their siblings, leading to redundant exploration of the same basins. In highly multimodal landscapes with narrow feasible regions, this redundancy becomes the dominant cost.

2.4 Method B: Elite-Resample Multi-Start (ER-MS)

2.4.1 Motivation and Interpretation

ER-MS augments MS by introducing a periodic selection–mutation operator that concentrates the worker population toward low-energy regions. Viewed abstractly, the R workers form an *interacting particle system* whose empirical distribution

$$\mu_k^R = \frac{1}{R} \sum_{r=1}^R \delta_{\mathbf{c}_k^{(r)}}$$

is driven toward π_T by alternating between independent SA evolution and population-level reweighting. This mirrors the structure of Sequential Monte Carlo (SMC) samplers, where a population of weighted particles is evolved and periodically resampled.

2.4.2 Resampling Operator

A resampling event is triggered every $K_{\text{resample}} = 200N$ MH steps. Workers are ranked by energy and the elite pool is defined as the top

$$k' = \lceil 0.25R \rceil$$

workers. Elite workers are preserved unchanged. Each non-elite worker r is replaced by a perturbed copy of a uniformly drawn elite r^* :

$$\mathbf{c}^{(r)} \leftarrow \mathbf{c}^{(r^*)} + \boldsymbol{\xi}_r, \quad r^* \sim \text{Uniform}\{1, \dots, k'\}.$$

2.4.3 Perturbation Model

The perturbation $\boldsymbol{\xi}_r$ is drawn independently for spatial and angular components. For polygon i , the displacement and rotation noise are

$$(\delta x_i, \delta y_i)^\top \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}(f)^2 I_2), \quad \delta \theta_i \sim \mathcal{N}(0, \sigma_\theta(f)^2),$$

where the standard deviations decay linearly with the elapsed fraction $f \in [0, 1]$ of the current probe:

$$\sigma_{\text{pos}}(f) = (0.02 - 0.015 f) L, \tag{2.1}$$

$$\sigma_\theta(f) = 5^\circ - 4^\circ f. \tag{2.2}$$

The full per-polygon perturbation covariance is therefore the block-diagonal matrix

$$\Sigma(f) = \text{diag}(\underbrace{\sigma_{\text{pos}}^2, \sigma_{\text{pos}}^2, \sigma_{\theta}^2}_{\text{polygon 1}}, \dots, \underbrace{\sigma_{\text{pos}}^2, \sigma_{\text{pos}}^2, \sigma_{\theta}^2}_{\text{polygon } N}).$$

Large early-probe noise allows workers to escape elite basins and explore new regions; decaying late-probe noise concentrates refinement near the current best solution.

2.4.4 Anti-Collapse Mechanism

Population collapse—all workers converging to the same local minimum—reduces ER-MS to a single chain and eliminates the benefit of parallelism. Collapse is detected via the normalized energy spread

$$\delta_{\text{rel}} = \frac{E_{\text{max}} - E_{\text{min}}}{\max(1, |E_{\text{mean}}|)} < 10^{-6},$$

evaluated at each resampling event. If this condition holds for two consecutive resamples, the perturbation scales are doubled for the immediately following resample cycle:

$$\sigma_{\text{pos}} \leftarrow 2\sigma_{\text{pos}}(f), \quad \sigma_{\theta} \leftarrow 2\sigma_{\theta}(f).$$

2.4.5 Relationship to SMC

In standard SMC, particles carry importance weights and resampling is performed proportional to those weights. ER-MS replaces importance weights with a hard rank cutoff (top k'), which is computationally cheaper but introduces selection bias: the retained configurations are not guaranteed to be representative samples from π_T . The mutation step (Gaussian perturbation) plays the role of the MCMC rejuvenation kernel in SMC, restoring diversity lost to selection.

2.5 Method C: Parallel Tempering (PT)

2.5.1 Replica Ensemble and Joint Distribution

PT maintains R replicas at a fixed geometric temperature ladder:

$$T_r = T_{\min} \left(\frac{T_{\max}}{T_{\min}} \right)^{(r-1)/(R-1)}, \quad r = 1, \dots, R,$$

so that $T_1 < T_2 < \dots < T_R$. The extended system targets the *joint* Gibbs distribution

$$\Pi(\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(R)}) = \prod_{r=1}^R \pi_{T_r}(\mathbf{c}^{(r)}).$$

The geometric ladder spacing gives a constant temperature ratio between adjacent rungs, which empirically yields uniform swap acceptance rates across the ladder.

2.5.2 Within-Replica Dynamics

Between swap events, each replica evolves independently under its own MH kernel targeting π_{T_r} :

$$\mathbf{c}_k^{(r)} \xrightarrow{P_k^{(r)}, T_r} \mathbf{c}_{k+1}^{(r)}.$$

2.5.3 Replica Exchange

Every $K_{\text{swap}} = 200N$ MH steps, an exchange is proposed between each adjacent pair $(r, r + 1)$. The swap acceptance probability is

$$\alpha_{\text{swap}}^{(r, r+1)} = \min\left(1, \exp\left[\left(\frac{1}{T_r} - \frac{1}{T_{r+1}}\right) \left(E(\mathbf{c}^{(r+1)}; L) - E(\mathbf{c}^{(r)}; L)\right)\right]\right).$$

Proposition 2.1 (Detailed Balance for Replica Exchange). *The swap acceptance probability $\alpha_{\text{swap}}^{(r, r+1)}$ satisfies detailed balance with respect to the joint distribution Π .*

Proof. Let $\mathbf{c}$ and $\mathbf{c}'$ be the ensemble states before and after swapping replicas r and $r + 1$. The ratio of joint probabilities is

$$\frac{\Pi(\mathbf{c}')}{\Pi(\mathbf{c})} = \frac{\pi_{T_r}(\mathbf{c}^{(r+1)}) \pi_{T_{r+1}}(\mathbf{c}^{(r)})}{\pi_{T_r}(\mathbf{c}^{(r)}) \pi_{T_{r+1}}(\mathbf{c}^{(r+1)})} = \exp\left[\left(\frac{1}{T_r} - \frac{1}{T_{r+1}}\right) \left(E^{(r+1)} - E^{(r)}\right)\right],$$

where $E^{(r)} = E(\mathbf{c}^{(r)}; L)$. The Metropolis acceptance ratio $\alpha_{\text{swap}}^{(r, r+1)} = \min(1, \Pi(\mathbf{c}')/\Pi(\mathbf{c}))$ then satisfies $\Pi(\mathbf{c}) \alpha_{\text{swap}}^{(r, r+1)} = \Pi(\mathbf{c}') \alpha_{\text{swap}}^{(r+1, r)}$ by the standard Metropolis argument. $\square$

2.5.4 Mechanism: Temperature-Assisted Basin Escape

High-temperature replicas flatten the effective energy landscape, allowing them to traverse barriers that would trap a cold chain. When a swap carries a favorable configuration from replica $r + 1$ to replica r , that configuration begins cold-chain refinement from a point already near a low-energy basin. This *temperature-assisted diffusion* is the core reason PT outperforms MS on landscapes with high, narrow barriers separating deep basins.

2.5.5 Early Stopping Criterion

Because our goal is feasibility rather than equilibrium sampling, a probe terminates only when the *coldest* replica ($r = 1$) reaches $E(\mathbf{c}^{(1)}; L) \leq \varepsilon_{\text{feas}}$. Feasible configurations visiting higher-temperature replicas are not accepted as solutions unless they propagate to replica 1 via an accepted swap chain.

2.5.6 Pilot Calibration

The temperature bounds $T_{\min}$ and $T_{\max}$ are set by the following pilot procedure:

1. Run a 10-minute pilot at $N = 100$, $R = 10$ with an initial temperature ladder.
2. Compute the empirical swap acceptance rate $\hat{\alpha}_r$ at each adjacent pair $(r, r + 1)$.
3. If $\hat{\alpha}_r \notin [0.15, 0.60]$ for any rung, multiply $T_{\max}$ or $T_{\min}$ by a factor of 2 and repeat.
4. Lock the calibrated values. Run a 5-minute sanity pilot at $N = 200$ to confirm validity at larger system size.

The target window $[0.15, 0.60]$ is a standard empirical criterion: below 0.15 the cold chain mixes too slowly; above 0.60 adjacent replicas are too similar for exchange to provide useful information flow.

2.6 Energy Landscape Interpretation

The relative performance of the three methods is determined by the structure of the energy landscape $E(\cdot; L)$. Three geometric features are particularly relevant:

- **Local minima.** Configurations $\mathbf{c}^*$ satisfying $\nabla E = 0$ with $E(\mathbf{c}^*) > 0$ are jammed states: partially overlapping packings with no locally improving move.
- **Energy barriers.** A barrier of height Δ between two basins requires the chain to traverse a region of energy at least Δ above the current state. Under the Metropolis rule, the probability of acceptance at temperature T scales as $e^{-\Delta/T}$, so barriers suppress transitions exponentially as $T \rightarrow 0$.
- **Basins of attraction.** Each local minimum is surrounded by a basin of configurations from which gradient-following dynamics converge to it. Wide, shallow basins are easily escaped at moderate temperatures; deep, narrow feasible basins require low temperatures to be refined once entered.

MS relies on independent discovery of distinct basins; it does not improve mixing within any single chain. ER-MS concentrates the worker population in the lowest-energy basins discovered so far, amplifying exploitation at the cost of diversity. PT enables direct inter-basin transitions by lifting configurations to temperatures where barriers are effectively invisible, then returning them to low temperature for refinement.

Chapter 3

Collision Detection for Non-Convex Polygons

Evaluating the energy function $E(\mathbf{c}; L)$ at every Markov chain step requires determining, for each pair of polygons, whether they overlap and by how much. This computation is the dominant cost kernel of the entire optimization: for N polygons, there are $\binom{N}{2} = O(N^2)$ candidate pairs, and each narrowphase test involves nontrivial geometric work. The collision system occupies the same structural role in this project as force evaluation in molecular dynamics or matrix multiplication in linear algebra—its throughput sets the ceiling on how many SA steps can be executed per second.

This chapter develops the mathematical foundations of collision detection for non-convex polygons, presents the three narrowphase implementations benchmarked in this study, and analyzes their cost structure.

3.1 Non-Convex Polygons and Convex Decomposition

The polygon used in this study is a planar non-convex (concave) polygon $\mathcal{P} \subset \mathbb{R}^2$ with V vertices. A polygon is convex if for every pair of interior points p, q , the segment $[p, q]$ lies entirely within $\mathcal{P}$. Non-convex polygons violate this condition; equivalently, their boundary is not everywhere locally on the same side of every edge normal.

The Separating Axis Theorem, which provides the most efficient exact intersection test, applies only to convex shapes. To handle non-convex polygons, we decompose $\mathcal{P}$ into a collection of convex *parts*:

$$\mathcal{P} = \bigcup_{j=1}^M C_j, \quad C_j \text{ convex},$$

where the parts cover $\mathcal{P}$ exactly (with at most boundary overlap). Two non-convex polygon instances $\mathcal{P}_i$ and $\mathcal{P}_k$ (at poses (x_i, y_i, θ_i) and (x_k, y_k, θ_k)) intersect if and only if at least one part

of $\mathcal{P}_i$ intersects at least one part of $\mathcal{P}_k$:

$$\mathcal{P}_i \cap \mathcal{P}_k \neq \emptyset \iff \exists j \in \{1, \dots, M\}, l \in \{1, \dots, M\} \text{ s.t. } C_j^{(i)} \cap C_l^{(k)} \neq \emptyset,$$

where $C_j^{(i)}$ denotes part j of polygon i transformed to world space. This reduces non-convex intersection to a collection of convex–convex tests.

Decomposition methods. Two decompositions are used in this study:

- **Triangulation (Method A).** $\mathcal{P}$ is decomposed into $M = V - 2$ triangles via fan triangulation from a reference vertex. Each part C_j is a triangle, giving the smallest possible number of vertices per part (3) at the cost of the largest number of parts.
- **Convex decomposition (Methods B and C).** $\mathcal{P}$ is decomposed into a smaller number of convex polygons, each with more vertices. Fewer parts reduce the number of part–part pairs; larger vertex counts increase the per-pair axis budget.

3.2 Coordinate Transforms

Each polygon instance i is defined by a pose $(x_i, y_i, \theta_i) \in \mathbb{R}^2 \times [0, 2\pi)$. Local-frame vertices $v \in \mathbb{R}^2$ are mapped to world frame by the rigid-body transform

$$T_i(v) = R(\theta_i)v + \mathbf{t}_i, \quad R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \mathbf{t}_i = (x_i, y_i)^\top.$$

The world-frame vertices of part $C_j^{(i)}$ are $\{T_i(v) : v \in C_j\}$. Edge normals transform under rotation only: if $\mathbf{n}$ is a unit outward normal in local frame, its world-frame counterpart is $R(\theta_i)\mathbf{n}$.

A naive implementation recomputes $R(\theta_i)$ and all transformed vertices inside the innermost pair loop, evaluating sin and cos once per pair–part test. The cached method (Section 3.7) amortizes this cost over all tests involving polygon i .

3.3 The Separating Axis Theorem

3.3.1 Statement and Proof Sketch

Theorem 3.1 (Separating Axis Theorem). *Let $P, Q \subset \mathbb{R}^2$ be compact convex sets. Then $P \cap Q = \emptyset$ if and only if there exists a unit vector $\mathbf{n} \in \mathbb{R}^2$ such that*

$$\max_{p \in P} \langle p, \mathbf{n} \rangle < \min_{q \in Q} \langle q, \mathbf{n} \rangle.$$

Such a vector is called a separating axis.

The “if” direction is immediate: if a separating axis exists then no point belongs to both projected intervals, so $P \cap Q = \emptyset$. The “only if” direction follows from the separating hyperplane theorem for convex sets: if $P \cap Q = \emptyset$, by convexity and compactness there exists a hyperplane (in $\mathbb{R}^2$, a line) strictly separating them, whose normal is the required $\mathbf{n}$.

3.3.2 Projection and Interval Test

For a convex polygon P with vertices $\{v_1, \dots, v_k\}$, the projection onto axis $\mathbf{n}$ yields the interval

$$I_P(\mathbf{n}) = \left[\min_{1 \leq j \leq k} \langle v_j, \mathbf{n} \rangle, \max_{1 \leq j \leq k} \langle v_j, \mathbf{n} \rangle \right].$$

Two intervals $I_P = [a, b]$ and $I_Q = [c, d]$ are *disjoint* if $b < c$ or $d < a$. Disjointness on any axis certifies non-intersection.

3.3.3 Candidate Axes for Polygons

For two convex polygons with k_1 and k_2 vertices respectively, it suffices to test only $k_1 + k_2$ candidate axes: the outward edge normals of each polygon. This follows from the observation that the extremal separating direction between two convex polygons must be perpendicular to a face of one of them.

Corollary 3.2. *Two convex polygons P (with k_1 edges) and Q (with k_2 edges) are disjoint if and only if a separating axis exists among the $k_1 + k_2$ outward edge normals of P and Q .*

The algorithm tests axes in sequence and returns DISJOINT as soon as any separating axis is found (*early-out*). If all $k_1 + k_2$ axes yield overlapping intervals, the algorithm returns INTERSECTING.

3.3.4 Penetration Depth

When the polygons do intersect, the minimum translational distance (MTD) required to separate them is the minimum interval overlap over all axes:

$$d_{\text{pen}} = \min_{\mathbf{n} \in \mathcal{A}} \max(0, \min(b_P, b_Q) - \max(a_P, a_Q)),$$

where $I_P(\mathbf{n}) = [a_P, b_P]$, $I_Q(\mathbf{n}) = [a_Q, b_Q]$, and $\mathcal{A}$ is the set of candidate axes. The MTD provides a natural penalty signal: deeper interpenetrations contribute more to the energy function.

3.4 Broadphase Filtering

For N polygons, there are $\binom{N}{2}$ instance pairs and $\binom{NM}{2}$ part pairs in the worst case. Narrowphase SAT is applied only to pairs that pass a cheap *broadphase* filter, which conservatively rejects pairs that cannot possibly intersect.

3.4.1 Axis-Aligned Bounding Box Test

The axis-aligned bounding box (AABB) of a transformed polygon instance $\mathcal{P}_i$ in world frame is

$$\text{AABB}(\mathcal{P}_i) = \left[\min_j (T_i(v_j))_x, \max_j (T_i(v_j))_x \right] \times \left[\min_j (T_i(v_j))_y, \max_j (T_i(v_j))_y \right].$$

Two polygons can intersect only if their AABBs overlap, since the AABB is a convex superset of the polygon. AABB overlap is tested in $O(1)$ via four scalar comparisons and rejects non-proximate pairs with no trigonometric cost.

Per-part AABBs are computed analogously for each convex part $C_j^{(i)}$, providing a second level of filtering before the narrowphase is invoked.

3.4.2 Spatial Hash Grid

For $N > 10$, the cost of testing all $\binom{N}{2}$ AABB pairs becomes significant. A spatial hash grid partitions the domain $[0, L]^2$ into cells of width h and assigns each polygon to the cells overlapping its AABB. Only polygons sharing at least one cell are candidate pairs.

The grid is stored as a compact hash table: cell keys are hashed to buckets, and each bucket stores a linked list of polygon indices. For a uniform random configuration at packing fraction ϕ , the expected number of neighbors per polygon is $O(1)$ for a cell size h proportional to the polygon diameter, giving $O(N)$ total candidate pairs and reducing the broadphase from $O(N^2)$ to $O(N)$ in expectation.

The grid is rebuilt every time a configuration is modified—captured in the `rebuild_derived` call—so its cost is amortized over all pair tests in the subsequent energy evaluation.

3.5 Narrowphase Method A: Triangle Decomposition

Method A decomposes each polygon into $M = V - 2$ triangles via fan triangulation from vertex v_1 :

$$\mathcal{P} = \bigcup_{j=2}^{V-1} \triangle(v_1, v_j, v_{j+1}).$$

Intersection between two polygon instances is tested by checking all $M^2 = (V - 2)^2$ triangle–triangle pairs.

Triangle–triangle test. Two triangles $\triangle ABC$ and $\triangle DEF$ intersect if and only if any edge of one separates them under SAT with 6 candidate axes (3 edge normals per triangle), or one triangle contains a vertex of the other. This is a fully degenerate instance of the general convex polygon SAT with $k_1 = k_2 = 3$.

Cost. Each triangle–triangle test requires at most 6 axis projections, each of cost $O(3)$ dot products, giving $O(18)$ floating-point operations per pair in the worst case with no early-out. The total narrowphase cost for one polygon–polygon test is $O(M^2) = O((V - 2)^2)$ in the worst case, with early-out reducing average cost substantially in non-intersecting configurations.

3.6 Narrowphase Method B: Baseline Convex SAT

Method B applies SAT directly to each pair of convex parts $C_j^{(i)}$ and $C_l^{(k)}$ in world coordinates. For each part pair, the algorithm:

1. Recomputes the world-frame vertices of $C_j^{(i)}$ by applying T_i to all local vertices.
2. Recomputes the world-frame edge normals of $C_j^{(i)}$ by applying $R(\theta_i)$ to all local normals.
3. Repeats steps 1–2 for $C_l^{(k)}$.
4. Tests all $k_j + k_l$ candidate axes via projection and interval overlap.

Steps 1–3 are performed fresh for every part pair, even when polygon i appears in multiple tests during the same energy evaluation. If polygon i has D neighbors, its transform T_i is applied $D \cdot M$ times: once per part per neighbor.

Cost. For a polygon with M parts each of k vertices, evaluating E_{overlap} for one polygon against all D neighbors costs

$$O(D \cdot M^2 \cdot k)$$

floating-point operations, where the M^2 factor counts part–part pairs and the k factor counts projections per axis. The trigonometric cost ($\sin$, $\cos$) is $O(D \cdot M)$.

3.7 Narrowphase Method C: Cached Convex SAT

Method C eliminates the redundant per-pair transform recomputations by caching world-frame data for all N polygons in a per-thread **Workspace** before the pair loop begins.

3.7.1 Pre-Pass: Cache Population

Before any pair test, a single pass over all N polygons computes and stores in the **Workspace**:

$$\text{World vertices: } v_{\text{world}}^{(i)} = T_i(v_{\text{local}}^{(i)}) \quad \forall i, \forall v, \quad (3.1)$$

$$\text{World normals: } \mathbf{n}_{\text{world}}^{(i)} = R(\theta_i) \mathbf{n}_{\text{local}}^{(i)} \quad \forall i, \forall \text{ edge}, \quad (3.2)$$

$$\text{AABBs: } \text{AABB}(C_j^{(i)}) \quad \forall i, \forall j. \quad (3.3)$$

The rotation $R(\theta_i)$ is evaluated once per polygon per energy evaluation, with $\sin(\theta_i)$ and $\cos(\theta_i)$ stored as scalars. All subsequent axis projections read from the cached arrays. Cache population costs $O(NMk)$ uniformly, amortized as $O(k/D)$ per pair test for a polygon with D neighbors.

3.7.2 Hint-Axis Pre-Filter

Before executing the full $k_j + k_l$ -axis SAT loop, Method C tests a single *hint axis*: the unit vector connecting the centroids of the two parts:

$$\mathbf{n}_{\text{hint}} = \frac{\bar{C}_j^{(i)} - \bar{C}_l^{(k)}}{\|\bar{C}_j^{(i)} - \bar{C}_l^{(k)}\|},$$

where $\bar{C}_j^{(i)}$ denotes the centroid of part j of instance i . If the projections of $C_j^{(i)}$ and $C_l^{(k)}$ onto $\mathbf{n}_{\text{hint}}$ are disjoint, the parts are immediately classified as non-intersecting at the cost of $2k$ dot products—substantially cheaper than the full SAT loop.

The hint axis is effective because separated polygons in a packing are most commonly separated along the direction between their centers. It provides a “free” early-out for the majority of non-intersecting pairs that survive AABB filtering.

3.7.3 Full SAT Fallback

If the hint axis does not separate the parts, the full SAT loop over all $k_j + k_l$ cached world normals is executed with early-out. Projections read directly from the pre-computed vertex arrays (3.1)–(3.2), avoiding all trigonometric recomputation.

3.7.4 Cost Reduction

Let D be the average number of polygon neighbors, M the parts per polygon, and k the vertices per part. Table 3.1 compares the operation counts per energy evaluation.

Table 3.1: Approximate floating-point operation counts per energy evaluation.

Step	Method B	Method C
sin/cos evaluations	$O(DNM)$	$O(N)$
Vertex transforms	$O(DNMk)$	$O(NMk)$ (pre-pass)
Axis projections (per pair)	$O(M^2k)$	$O(k)$ hint + $O(M^2k)$ fallback

For $D \gg 1$ (dense packing) the trigonometric saving is the dominant effect: Method C evaluates sin/cos exactly N times per energy call regardless of density, while Method B evaluates them DNM times.

3.8 Connection to the Energy Function

The collision system feeds directly into the energy function introduced in Chapter 1:

$$E(\mathbf{c}; L) = E_{\text{overlap}}(\mathbf{c}) + E_{\text{boundary}}(\mathbf{c}; L).$$

Overlap penalty. For each pair of polygon instances (i, k) found to intersect by the narrow-phase, the overlap penalty is the sum of penetration depths over all intersecting part pairs:

$$E_{\text{overlap}}(\mathbf{c}) = \sum_{i < k} \sum_{\substack{j, l \\ C_j^{(i)} \cap C_l^{(k)} \neq \emptyset}} d_{\text{pen}}(C_j^{(i)}, C_l^{(k)}).$$

This penalty is zero if and only if no two polygons overlap, i.e., the configuration is collision-free.

Boundary penalty. For each polygon i , the boundary penalty accumulates for every vertex lying outside the container $[0, L]^2$:

$$E_{\text{boundary}}(\mathbf{c}; L) = \sum_{i=1}^N \sum_{v \in \mathcal{P}_i} \|\max(\mathbf{0}, -T_i(v))\| + \|\max(\mathbf{0}, T_i(v) - L\mathbf{1})\|,$$

where the max is taken componentwise. A configuration is feasible (Section ??) when both penalties are below the threshold $\varepsilon_{\text{feas}}$.

3.9 Benchmark Design

3.9.1 Objective

The microbenchmark isolates the collision system from the SA loop to measure throughput in controlled conditions. Its purpose is twofold: (i) to verify that Method C achieves a concrete speedup over Methods A and B before committing to the 4-hour hero runs, and (ii) to quantify the rejection efficiency of the broadphase and hint-axis filter, which determines what fraction of SA steps incur full SAT cost.

3.9.2 Protocol

A fixed set of N polygon instances is placed at random non-overlapping positions. The full $\binom{N}{2}$ pair evaluation loop is then executed R_{bench} times, and wall-clock time is recorded. The same configuration is used for all three methods, so differences in timing reflect only the collision algorithm.

The benchmark is invoked as:

```
./bench <N> <R_bench> <out.csv>
```

3.9.3 Primary Metrics

The following quantities are recorded for each method:

Table 3.2: Benchmark metrics collected per method.

Metric	Definition
Pairs total	$\binom{N}{2}$ instance pairs evaluated
Broadphase pass	Pairs surviving instance AABB filter
Narrow calls	Narrowphase invocations
Collisions	Intersecting pairs found
Wall time (s)	Total elapsed time
Mpairs/s	Million instance pairs per second

For Method C, additional counters are recorded to expose the filtering cascade:

Table 3.3: Method C internal diagnostic metrics.

Metric	Definition
Part pairs	Total part-part pairs entering narrowphase
Part AABB pass	Part pairs surviving per-part AABB filter
Hint tests	Hint-axis projections executed
Hint rejects	Pairs rejected by hint axis
SAT full calls	Full SAT loop invocations
SAT hits	SAT calls returning intersection
SAT axes tested	Total axes evaluated across SAT calls
Avg axes / SAT call	SAT axes tested $\div$ SAT full calls

The ratio **Hint rejects** / **Part AABB pass** measures the effectiveness of the hint-axis pre-filter: a high ratio indicates that most surviving AABB pairs are resolved cheaply without entering the full SAT loop. The **Avg axes / SAT call** metric quantifies early-out effectiveness within the SAT loop itself.

3.9.4 Scaling Study

The benchmark is run at $N \in \{10, 20, 50, 100, 200\}$ to characterize how throughput scales with system size. For each N , three placement regimes are evaluated:

1. **Sparse:** polygons placed at random with low overlap probability. Tests broadphase efficiency.
2. **Dense:** polygons packed at high density near feasibility. Tests narrowphase cost in the regime the SA spends most time.
3. **Jammed:** polygons at a jammed configuration with maximal neighbor count. Tests worst-case narrowphase cost.

3.9.5 Results and Interpretation

Method	Pairs total	Broad pass	Narrow calls	Collisions	Time (s)
Triangle (A)	-	-	-	-	-
SAT baseline (B)	-	-	-	-	-
SAT cached (C)	-	-	-	-	-

Method C diagnostic	Value
Part pairs	-
Part AABB pass	-
Hint tests	-
Hint rejects	-
SAT full calls	-
SAT hits	-
SAT axes tested	-
Avg axes / SAT call	-

The expected outcome, borne out by the cascade structure of Method C, is as follows. Most polygon pairs are rejected by the instance-level AABB filter, so the broadphase pass rate should be small ($< 20\%$ for sparse configurations). Of the pairs that survive, the per-part AABB filter and hint-axis pre-filter together reject the majority before any full SAT loop is invoked. The full SAT loop is called only for pairs that are either intersecting or geometrically close to intersection, and among those calls, the early-out terminates the axis loop after on average far fewer than $k_j + k_l$ tests. This cascade is the mechanism by which Method C achieves substantially higher throughput than Methods A and B at equal correctness.

3.10 Integration into the SA Loop

In the context of the optimization study, the collision system is invoked at every accepted SA proposal that changes any polygon’s pose. Per-thread `Workspace` instances are allocated once at thread initialization and reused across all energy evaluations, avoiding heap allocation in the inner loop.

After a pose update, `rebuild_derived` recomputes the world vertices, rotated axes, and AABBs for the moved polygon only, and updates its spatial hash cell memberships. A full grid rebuild is triggered only when global configuration changes require it (warm-start, clone, or swap operations). This incremental update strategy reduces the pre-pass cost of Method C from $O(NMk)$ to $O(Mk)$ per SA step in the common case where only one polygon moves at a time.

The collision throughput therefore sets the wall-clock budget for SA step rate: if Method C achieves T_{narrow} microseconds per narrowphase call and the average polygon has D neighbors at

M parts each, the expected energy evaluation time per SA step is approximately $D \cdot M^2 \cdot T_{\text{narrow}}$ microseconds, plus the fixed overhead of the AABB and hint-axis filters. Chapter 4 uses the benchmark throughput numbers from Table ?? to project feasible SA step rates for each problem size N .

Bibliography

- [1] Bruce Hajek. Cooling schedules for optimal annealing. *Mathematics of Operations Research*, 13(2):311–329, 1988. doi: 10.1287/moor.13.2.311.